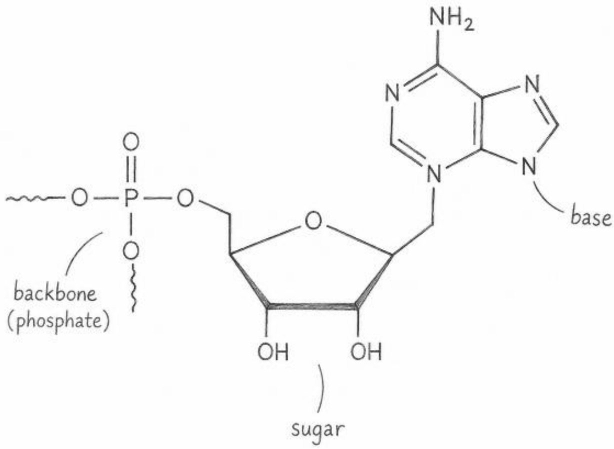
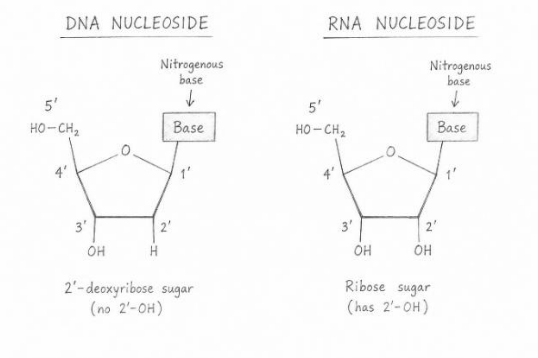


Nucleic acids are polymers composed of repeating nucleotide units, each consisting of a nitrogenous base, a five-carbon sugar, and a phosphate group. In DNA, the sugar is 2′-deoxyribose, whereas RNA contains ribose, which possesses an additional hydroxyl group at the 2′ position. Nucleotides are connected through 3′→5′ phosphodiester linkages to form a sugar-phosphate backbone, and the sequence of bases (adenine [A], guanine [G], cytosine [C], thymine [T] in DNA, and uracil [U] in RNA) encodes biological information. Understanding nucleic acid therapeutic chemistry requires familiarity with nucleotide nomenclature and numbering conventions, particularly the sugar positions 2′, which is a common site for chemical modification. ASOs frequently incorporate modifications to the phosphate backbone, such as phosphorothioate (PS) linkages in which a non-bridging oxygen is replaced by sulfur, or modifications to the sugar moiety, including 2′-O-methyl (2′-OMe), 2′-O-methoxyethyl (2′-MOE), locked nucleic acids (LNAs), and constrained ethyl (cEt) nucleotides. Base modifications may also be introduced to improve stability, affinity, or specificity.



DNA is composed of repeating nucleosides that are connected by phosphate groups to form a linear polymer. Each nucleoside consists of a nitrogenous base attached to a five-carbon sugar called 2′-deoxyribose. The carbons of the sugar are numbered 1′ through 5′, and these numbers are essential for describing nucleic acid structure and chemistry. The 1′ carbon is attached to the base, the 2′ carbon contains a hydrogen atom in DNA (rather than a hydroxyl group as in RNA), the 3′ carbon contains a hydroxyl (–OH) group, and the 5′ carbon is attached to a phosphate group. During DNA synthesis, the phosphate group forms a phosphodiester linkage between the 3′ hydroxyl of one nucleoside and the 5′ position of the next nucleoside, creating the characteristic 3′→5′ sugar-phosphate backbone. These numbering conventions define the directionality of a DNA strand, with one end designated the 5′ end and the other the 3′ end and provide the framework for understanding many nucleic acid modifications used in antisense oligonucleotides and other nucleic acid therapeutics.